

Critical Moments: Supporting Homeless Journeys through Design

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ABSTRACT

Individuals facing homelessness navigate through different moments and situations that bring specific challenges to their lives. A thorough investigation of their 'journeys' through such a difficult experience can provide new opportunities about how technologies can support individuals through homelessness. Based on a qualitative interview study involving nine individuals struggling with homelessness in an Australian city, we aimed to understand and trace critical moments of their struggles through a 'journey tracking' exercise. Our findings unpacked challenges and everyday resilient practices surrounding homelessness and the role technology can play in it. We discuss implications around developing services to provide emergency support to the homeless population based on our findings.

CCS CONCEPTS

• Human-centered computing; • Human Computer Interaction.;

KEYWORDS

Homelessness, Design, Qualitative Research

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1 INTRODUCTION

"Homelessness is one of the least understood social issues. The public image of homelessness and public perceptions of the nature and causes of homelessness have little relation of the situation" [10].

Homelessness is one of the growing problems in all the countries around the world. According to a recent report, about 2 percent of the world's population may be homeless and another 20 percent lacks adequate housing [20]. A homeless person is an individual

who seeks primary, night-time residence at publicly or privately-operated shelter homes providing temporary living accommodation [21]. Homelessness is not limited to housing problems only, it is as much about individuals facing a lack of connection with friends, family, society, and a lack of control over personal matters. Homelessness increases exposure to unclean and unsafe living conditions that adversely affect the health and wellbeing of individuals [25].

Homelessness is categorized as a phase in an individual's life that can be short-term or ongoing [21]. Chamberlain and Mackenzie [5] have classified homelessness in three categories: Primary Homelessness (rough sleepers with no adequate sleeping arrangements), Secondary Homelessness (living in temporary shelters, or couch-surfing), and Tertiary Homelessness (living in boarding rooms and caravan parks). Individuals navigate through these categories of homeless, as their situation changes, for better or for worse.

In this paper, we present findings from an interview-based study which was aimed to study critical moments in the journey of homelessness. In particular, we aimed to explore and understand the problems faced by the homeless community in navigating from one place to another and the different stages of their homelessness. We conducted qualitative, contextual interviews with nine participants in a metropolitan city in Australia. We used a "journey tracking" exercise which enabled participants to draw their homelessness journeys on a piece of paper. Through this, we were able to highlight critical moments in individuals' lives that have contributed to their homelessness. Our findings highlight the most common causes of homelessness, measures taken by participants to combat their situation, and common traits and techniques followed by them. Based on our findings, we provide design implications to support individuals' journeys through their homelessness experiences. We believe that while technology may not be able to solve such a deeply rooted problem, it can certainly support people navigate through their homelessness experiences.

2 RELATED WORK

A person who lacks a fixed, regular, and adequate night-time residence and seeks a primary night-time residence is considered homeless [12].

There are 1.6 billion people who have inadequate housing that is around 20% of the total world population [2]. This is growing every day. There has been a rise in homelessness among young people aged 13-30, resulting in adverse social and economic consequences. Different countries have different definitions of homelessness [11]. In Australia, individuals are considered homeless if they do not have alternative accommodation and their current stay provides insufficient night-time shelter. The USA has a population of 550,000 homeless people; however, it has been argued that the actual figure might be more than quoted [26]. Nearly 1.56 million individuals in

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the USA have accessed emergency shelter homes at any point in time [1]. Studies reveal that homelessness can be caused by various reasons such as family violence, losing a job or unemployment, drug abuse and alcohol addiction [4], feeling unsafe at home, family and relationship problem, physical, and mental health issues and falling into situations where one can no longer pay rent to afford a house [22].

Technology plays a crucial role in providing social support to homeless people. A study has found that 44% of homeless individuals use mobile phones to stay connected with their friends, family, and service providers [12]. Homeless young people are adopting digital technologies quickly. Most of them can afford a mobile phone, which helps to maintain their ties with family through social media platforms like Facebook [7]. Usage of mobile phones has educated the homeless community. It has also provided them safety from the cruel environment and helps them be safe [9]. The research was done on how mobile phones have helped in unsafe situations and what improvements need to be done in this area using methods like value sketches, written value scenarios, group discussions, and interviews [19]. Shelter homes provide Internet access and computer access to homeless people. Technology not only helps them in connecting to their family but also helps them in learning educational programs and finding employment. This boosts their self-esteem and helps them feel like to be a part of society [18].

Uplifting the morale of homeless individuals is important. A study report suggests that music plays a significant role in it [14]. The study was conducted in shelter homes for building interest in music among the homeless community and was provided musical training [16]. With few years of training, it was noticed that the people have attained self-confidence. By perceiving the successful results, more focus was done on verbal and design activities like drawing, painting, and writing. Technology helps people from all works of life. The homeless community is one of them.

A research study revealed that how technology including mobile phone, bus passes, computers, etc. affect the daily lives of the homeless people and ample of opportunities for technological intervention to help this community [8]. There is a huge need to re-test the assumptions made in HCI in this domain and closely work on the relationship of people with technology. Le Dantec has outlined the need to examine different techniques for finding appropriate technology to be used in homes [6]. The need for HCI technology in a social environment is required now more than ever. By interviewing homeless people about how they perceive technology and how it is relevant to them, it was found out that there is a plethora of design innovations which is required for the homeless community and can be done effectively by using HCI technologies [27].

3 STUDY

3.1 Methodology

We engaged with a homeless shelter to first build rapport with its members and gain an understanding of the existing services provided by the shelter. The shelter was recently established and could only provide accommodation support for up to eight individuals. As we started engaging with the shelter members, we realized that several of the members were often skeptical of our intentions. The shelter's manager advised us that, "these people have told their

stories so many times to many different people that they may not be interested in telling their stories again." We realized that to understand their story, we needed to be a part of their community. The first author volunteered for various activities organized by the shelter and worked closely with other members. Through these volunteering opportunities, we were able to meet our nine participants (three from the shelter itself) who were willing to share their stories for our research.

Interacting with homeless communities to learn about their experiences and life stories has high risk and major ethical considerations. Our project had acquired ethical clearance from our institution. The ethics specified that participants will be interviewed in spaces where they would feel more comfortable. It also included the voluntary nature of the study (with a possibility to opt-out at any stage), de-identification of data, and minimal collection of personal data. Participants could choose not to answer any question that may be difficult for them. We were fully aware of the imbalance of power between ourselves and participants. The first authors' efforts in building rapport with participants ensured to minimize a sense of authority, social and political distance that comes from being a researcher.

Our interviews lasted for 45-60 minutes with each participant. The interview process began with questions related to their day-to-day life, their education and family background, their use of technologies, and more importantly listening to their stories around how they navigated through their homelessness struggles. To make our interview process more engaging, we employed a "journey tracking" activity as a part of our interview. Similar elicitation methods have been used when engaging with vulnerable populations [13]. In this exercise, participants were asked to draw a map of their journey so far using the drawing material we provided. This mapping helped us identify the common habits and traits adapted by people at different stages of homelessness. This provided a comprehensive picture of their life. Various questions were asked related to their journey to understand the role of technology in each phase of their journey and how it had helped them. The drawings produced during the journey tracing activity helped us guide our questions. We gave \$20 gift vouchers to all our participants as a token of appreciation. The interviews were audio-recorded while a few hand-written notes were also captured. The interviews were transcribed later and analyzed using the thematic analysis [3] technique. Here, the authors started by reading through the interview transcripts thoroughly, followed by applying coding based on recurring themes on specific instances in our participants' lives. We developed larger categories and themes that inductively emerged from our coding. Following this, we developed three themes around problems and challenges, participants' techniques to cope with these problems, and their ongoing efforts to bounce back. These themes will be discussed further in our findings section.

3.2 Participants

We were able to recruit three participants from the shelter that we initially engaged with. Following our interactions with some of the members at the shelter, we were able to reach out to six more participants through flyers and word of mouth. Out of the nine participants, five were female and four were male. Table 1 provides

Table 1: Details of the Participants

#	Gender	Homelessness	Age	Causes
P1	Female	Secondary	35-40	Partner's Death, Alcohol Abuse
P2	Male	Tertiary	50-55	Partner's Death, Drug Abuse, Gambling
P3	Female	Secondary	25-30	Domestic Violence
P4	Female	Tertiary	25-30	Parents Death
P5	Female	Tertiary	18-25	Parents did not concur with her sexuality
P6	Male	Tertiary	18-25	Drug abuse
P7	Female	Secondary	15-20	Suffered homelessness at 15 yrs. Unfit Parents
P8	Male	Tertiary	25-30	Mental illness, Partner's Death
P9	Male	Tertiary	35-40	Failed business

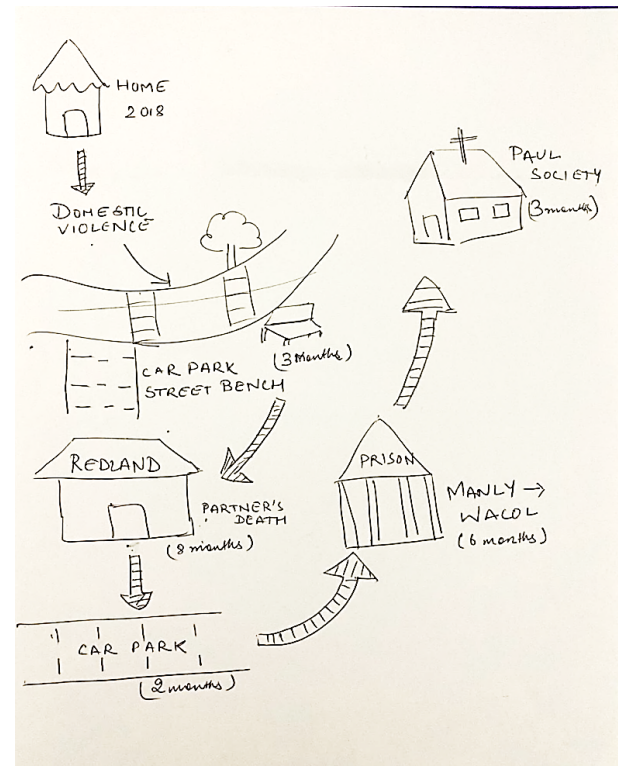
a detailed description of our participants. At the time of the study, none of the participants were living on the streets. They were either living in sheltered houses or couch surfing with their friends.

4 FINDINGS

Through interviews and journey tracking exercises, we found that the participants used a varied set of tactics and techniques to deal with their problems while navigating through different phases of life. In the following, we will start with P3's journey tracking activity and describe how her journey has been like.

P3 drew her journey where we get detailed insights into different phases of life, as shown in Figure 1. This map clearly outlines the reason for being homeless and her journey from being in the primary homelessness phase to tertiary and back to primary and finally to secondary homelessness phase. This map depicts how she survived on the streets and her struggles through those phases. She started her account by stating that after her marriage she was prone to domestic violence from her husband since 1st month of marriage. One day her husband came drunk and started beating her with an iron rod, set her clothes on fire, and sprayed petrol on her – pushing her to fire. She ran away from her marriage. Following that, she wandered around on streets away from her husband's terror and hiding at less crowded spots to survive. She found a partner after 3 months of living on the streets and moved in with him in a locality called Redland for 8 months. At that point, things were coming together for her and the couple was planning on getting married – until her partner died in an accident in a motorway car accident. She was taken aback and was devastated. She was back on the streets – primary homelessness. She got into illegal activities, slept in different car parks for 2 months, and was arrested because she stole someone's laptop and sold it to make money. She was in a prison for 6 months.

After getting released from jail, she met a lady from a Church who offered her to stay at a shelter home run by the St. Paul's Society. At the time of our interview, she has been staying there for the last 3 months. She found herself at peace as she started getting involved in religious activities at the church while helping out the church in cleaning and maintaining. She has also started knitting with other women in the church. It gives her immense pleasure and keeps her mind diverted from her horrendous past. She talked about using mobile phones while she was sleeping rough to find opportunities to make money. Later she started using the internet to learn skills and watch movies while staying at Paul's society.

**Figure 1: Journey tracking of Participant 3**

4.1 Social Aspects

People struggling with homelessness are vulnerable to safety issues. They are often scared of getting robbed or mugged. Female participants indicated that they felt scared of getting assaulted during this state. All of our participants who faced primary homelessness in the past mentioned that they preferred staying and roaming in groups. It made them feel safer and stronger. P1 mentioned that when she was living rough, she moved in groups and slept in groups in any location. She further explained that one older man used to sleep with them to protect them at night. This small act of kindness by the older man made them feel safe. She further commented, "It felt like we were in it together and fighting together gave us strength." Staying in groups was a very helpful strategy

for people who were experiencing homelessness for the first time. With the help from others in a group, the newcomers would get to know about restaurants, different places, and what time they could get free food from a different charity organization.

Sharing knowledge with friends and looking after one another was another common strategy we saw in our participants. P8 used Orange Sky to do his laundry. Orange Sky [24] is a mobile laundromat where a set of volunteers would help homeless individuals with their laundry. He talked about sharing the information about this service among his friends who lived on the streets. Following this, P8 and his friends used to come together and do laundry on the same days. Even during the hard times, this provided a sense of togetherness and a day to look forward to. P7 who was facing secondary homelessness (she lived with other female members in a female-only shelter) mentioned that “we have rosters for kitchen and cleaning duties. This encourages us to live a normal lifestyle with others under the same roof”. She further quoted that, “It is so amazing to see how few people in the house have come a long way from being locked in the room to getting up and making a cup of coffee for themselves”. P7’s shelter house regularly hosted garage sales. The residents in the shelter house actively participated in such activities, which helped them organize and open the sale to the general public. P7 mentioned that she looked forward to a garage sale where she could organize the event and interact with people. This made her lifestyle “normal and happy”.

4.2 Technological Aspect

Most of our participants struggled with accessing technologies such as laptops, mobile phones, chargers while facing primary homelessness. We learned that while it was important for all the participants to own and use such technologies, it made them vulnerable as potential targets of getting robbed. P4 commented, “Finding charging points was a big struggle. But it became worse when my charger was stolen and had to go to stores to ask if I could charge my phone for a while. Sometimes I was refused because I was not dressed properly.” In the case of P1, in her current shelter house, she was forbidden to have access to any mobile and communication technologies as she had severe substance abuse problems. She was only able to use the shelter’s landline phone that was monitored by the shelter officials. P1 mentioned that even when she is on her path to recovery, she would not have direct access to mobile phones.

Participants heavily used social media platforms like Facebook and Messenger to contact their friends and family. Few participants joined various homeless support groups on Facebook. This helped them connect to the larger community and provided a platform to look for jobs, seek help in terms of accommodation or any monetary help. These groups often posted information about restaurants offering free food or any kind of free services for the homeless. Participant P4 commented “I used my mobile phones to surf and browse for entertainment. I actively started looking for jobs on Seek, Facebook and posted about myself. I came forward and wrote my story on Facebook’s job portal and highlighting that I am looking for a job. One day I got a call from a dumpling shop to ask me if I can join immediately as a waitress.”

Participants often kept on looking for free Wi-Fi around the city so that they can access the services available on the internet.

Few of the participants even used to go to the city library to avail these services. Participants mentioned that they often went to the library to make their resumes and printed it from the library. Many participants used government welfare agency – Centrelink’s concession cards that can be used for travel and health care.

4.3 Navigational Aspects

The journey tracking exercise clearly showed how our participants navigated through different phases of their life. Participants build routines around conducting their everyday activities such as showers, laundry, use of toilets, and so on. They would go to public toilets available in council parks, go to city libraries to use computers, be present at specific places where free food and soup kitchens are scheduled. One of the strategies that came out quite clearly through several participants was that participants chose to go around different places and never made a fixed spot. P8 discussed that he struggled to live on his own due to his mental health issues after his partner’s death. He first moved into his father’s place but was back on the streets in less than 6 months. He described how he wandered around different streets to survive. He commented that he “slept on beaches by hiding from lifeguards and camped around places to stay. It was becoming impossible to carry so many things with me so I only took essentials in one bag pack and left the rest. Carried my documents, mobile phone, laptop, some cash, bus card, clothes, thongs, jumper, shoes, and a bottle of water – travel lighter.” There was always a fear of getting caught in residents’ eyes, hence trying different locations was an important strategy. In a different example, P2 mentioned that when he became homeless and was forced to be on roads, “I carried some clothes, jumper, shoes, and phone. I also carried my identity card and Medicare card. Sleeping on roads was not easy, often begged on streets, smoked weed with a gang and squatted on the streets later at night.” These two participants highlight the importance of traveling light with essentials required during navigation. They even highlighted how impractical it became to carry stuff around and always being scared of losing them and getting robbed. This technique helped their movement from one place to another easier.

The use of technologies was seen to be very important for our participants. P9 who lived in his car for quite a while mentioned that he used Google Maps to find out different locations where he could take shower and park his car for free. P9 highlighted how finding safe to sleep was the biggest challenge for him. He stated “living in a car with wife, kids, and a dog was the most difficult time. I drove kids to school and when I ran out of fuel I had to find a place near to school so that I could walk them home. Finding free parking overnight and during the day was a challenge because everything was paid and I did not want to park at dodgy locations as I had kids. I often looked for places with free Wi-Fi and good cellular connection where I could call people for help.”

5 DISCUSSION & FUTURE WORK

Our study aimed to understand the journeys of homeless individuals so that we can unpack problems and challenges that linger around homelessness. While Woelfer et al. [15-18] and Le Dantec’s [7] highlighted the role technologies play in supporting homeless individuals, our work takes a step back and looks into the navigational

aspects of homelessness. The use of journey tracking activity while interviewing participants highlights and compares how specific decisions taken by an individual at each step of life affected their life and lead them to different paths. This study also demonstrated specific strategies used by homeless individuals for survival. All the nine participants have used the internet while facing homelessness. In the case of P1 who was forbidden to carry mobile phones, regularly visited the city library to use the internet. Through the study conducted, it was noted that usage of digital technologies helped the young homeless individuals (such as participants 5, 6, 7, and 9) to get employment and educational opportunities quickly and they were able to overcome few issues of homelessness quickly making their situation better. Being connected with friends and family provides good physical and mental health. Le Dantec found the usage of mobile phones very common among the homeless community [7]. Our findings aligned with it, and highlighted how people facing different levels of homelessness rely on mobile phones to search, communicate, search for housing, job opportunities, know about services available for this community. In the following, we provide some preliminary ideas for technology design.

The sheltered houses generally focus on providing basic amenities like food and housing. They can strengthen the lives of individuals by focusing more specifically on individuals and their specific situations. They could incorporate confidence-boosting sessions where residents can interact with experts to learn about how to keep their morals high. Shelter providers frequently face funding problems and a lack of access to technologies through which they can keep records of individuals' performances digitally and track their progress.

Homeless individuals face challenges in carrying their luggage while navigating through different places. Often, they struggle to prioritize what to carry and they are highly prone to theft. In such cases, carrying essential documents for getting a job gets misplaced. This makes it very difficult for them to get a job and become self-sufficient. This problem can be addressed by providing physical lockers in a different locations where homeless individuals could come and store their important documents. Postal and courier services do offer lockers and storage spaces with a fee that may not be affordable to a majority of homeless individuals. A service where struggling individuals can avail themselves such physical storage spaces will not only protect important documents and other objects but will make individuals' navigation easier where they would not be worrying about the safety of their personal belongings. Furthermore, it will help them protect their privacy and grant easy access to them anytime.

5.1 Future Work

We found that our participants used several tactics to find free food, safe places to sleep, and free housings. These may not be completely alleviated by technologies, but they can provide support to homeless individuals navigate through their homelessness journeys. Our future work involves building on the existing tactics of homeless individuals. In the following, we list potential areas where technology can play role in the betterment of homeless individuals.

- **Connect people facing similar problems.** Being alone in the journey while moving from one place to another makes

one vulnerable and lonely. While interviewing our participants, we drew an interesting insight into how they found it useful to roam around in the company of others in a similar situation. This gave them safety, confidence, and a feeling of togetherness. An application that connects people who might be at different stages of homelessness will help. Applications can also act like an integrated platform where exclusive information and help required for this community can be posted. It can act as a single platform dedicated for the homeless community to post updates about jobs, living arrangements, and so on facilitating a new social media platform built exclusively for his community.

- **Locating free services nearby.** Being homeless, sleeping rough on streets, and carrying limited luggage while traveling from one place to another to survive, makes homeless individuals in dire need of clean clothes. Location-based technologies can be explored to support individuals to share information about the places that are providing free services around food, clothing, and washing. A lot of restaurants donate their excess food to the homeless population. Some directly serve them after their closing hours while some partner with other services to reach this community [23]. There are exclusive soup kitchens for homeless people where they can avail free food. Such services can be grouped in the application to keep the community informed about such offerings.
- **Locating safe places.** In Australia over 8,000 people on average sleep rough every night in street, on benches, in parks, and under bridges [1]. Homeless individuals often suffer from sleep deprivation due to extreme conditions on the streets. Sleeping in an unsecured place makes one quite anxious and scared of being attacked and robbed and hence one does not get peace of mind. By using a location-based service, homeless individuals can be aware of such places and even share such information with others.

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